

Eight Things Cars Still Get Wrong

This notebook is design research, not a certification. Nothing here has been crash-tested, homologated, or built. It is a record of eight places where vehicle design keeps making the same trade, and the direction C&W is taking instead. You can use it to look at any vehicle - yours, one in a car park, one in a showroom - and see things you did not see before.

How to read a gap

A gap is not a defect. A defect is something that went wrong. A gap is something that was traded away on purpose, usually for a good reason, and then never revisited after the reason changed.

Every entry below has the same three parts. The baseline is what is broadly true of vehicles today. The direction is what C&W is designing toward instead. The honest state is what is still unproven - and it is printed in the same size as everything else, because a research notebook that hides its unknowns is marketing.

1. The occupant is packaged last

Baseline: a performance vehicle's proportions are usually settled first, and the people go in afterwards. Comfort, entry, exit and visibility get what room is left over once the silhouette is decided.

Direction: settle the occupant envelope first. Comfort, egress, visibility and survivability define the package, and the surfacing happens around them. You cannot recover cabin space late; you can restyle a roofline late.

Honest state: open. The anthropometric target population is not locked, and until it is, "occupant-first" is a principle rather than a dimension.

2. There is no room in a side impact

Baseline: a frontal impact has a whole engine bay to crush through. A side impact has a door card, a few inches of structure, and then a person. That asymmetry is geometric, not a matter of effort.

Direction: increase the usable side crash volume and give the load somewhere to go instead of into the occupant.

Honest state: open, and one idea inside it is unvalidated. A deployable external shield - structure that

moves outward before contact - is an appealing concept with no evidence behind it yet. It is written down as research, not as a feature.

3. Water defeats the electrics before it defeats the car

Baseline: in a submersion, the failure that traps people is often electrical. Windows and latches stop answering long before the structure gives up.

Direction: mechanical egress that does not depend on power, electrical isolation on water entry, and beaconing so the outside world knows where the vehicle is.

Honest state: open, with two claims explicitly unvalidated - a days-long breathing reserve, and extracting oxygen from surrounding water. Both are recorded as research questions. Neither is a capability, and neither should be described as one.

4. Tall vehicles are shaped to catch people

Baseline: high ground clearance creates two problems at once. A smaller vehicle can go under a larger one, and a pedestrian meets a flat vertical face at chest height instead of a sloping one at hip height.

Direction: keep the high seating people actually want, but control the lower envelope - smooth anti-hook geometry, sensing, and speed management where the conflict is most likely.

Honest state: open. It needs a biomechanics and clearance trade study, because this is a case where the two goals genuinely pull against each other and the answer is a number, not an opinion.

5. Repair was never a design input

Baseline: assemblies get more integrated every year. A part that used to unbolt now comes bonded to three others, and a fault that used to be visible now lives in software.

Direction: modular service zones, crash modules that can be safely replaced, diagnostics that show evidence rather than a code, documented part provenance, and repair instructions treated as part of the product.

Honest state: this is the only one of the eight with no listed blocker. It is not waiting on a trial or a study - it is waiting on being designed. Repair access has to be decided before the styling freeze, because after the freeze it is not a decision any more.

6. Wheels got big for the photograph

Baseline: large wheels with thin tyres look decisive in a press image. They also add unsprung mass, raise cost, and turn an ordinary pothole into a bent rim.

Direction: an original deep-dish wheel language - the VYC 2ST family - validated against fatigue, impact, mass, brake cooling and real tyre loads before it is drawn as a shape.

Honest state: open. The final wheel geometry has not been designed. This notebook therefore contains no picture of it, and that is deliberate: at C&W, no image is made before the geometry gate passes.

7. Door handles solved the wrong problem

Baseline: flush and electronic handles buy a small aerodynamic gain and a clean surface. They cost discoverability - a stranger, a child, or a firefighter has to work out how to open the door, sometimes without power.

Direction: an ambidextrous rapid-access interface with a mechanical fallback that does not need to be explained, on a reinforced hinge and latch architecture.

Honest state: open, pending human-factors trials. Nobody can tell you from a desk whether a handle is discoverable under stress. That is measured, or it is guessed.

8. After the crash, the system falls apart

Baseline: the vehicle knows it crashed. The road does not. The dispatcher learns second-hand. The rescue crew arrives without knowing how many people are inside or what the battery is doing.

Direction: connect vehicle, road corridor, dispatch and rescue, and pass along a confidence level with every claim rather than an assertion - with privacy and due-process gates built in, not added after.

Honest state: open, and the blockers here are not engineering. They are jurisdiction, dispatch authority and response staffing. This is the entry on the list that money and cleverness alone will not close.

The rule underneath all eight

Systems may operate independently, but they must stay mutually observable. A failure should isolate without blinding the whole vehicle. Repair access is designed before the styling freeze. Common service items are reachable without taking unrelated things apart.

That is the C&W Safe and Serviceable architecture in four sentences. It is a specification in progress, not a finished standard, and it is the reason this notebook keeps saying open.

Take it outside

Pick any vehicle you can walk up to and answer these, out loud, in order.

Where does a side impact put its load? Follow the line with your finger. If you cannot find it, that is the finding.

If the power died right now, how would someone outside open this door? How long did it take you to work it out?

Which service item is reachable without removing something unrelated? Name one. Then name a second.

Stand where a child stands. What can the driver not see?

What on this vehicle exists for the photograph rather than for the person inside it?

You will not get a score. You will get the habit, and the habit is the product.

What this notebook is not

It is not a safety certification. It is not test data. It is not a claim that any C&W vehicle exists in physical form, has been built, or has been validated by anyone. Every one of the eight gaps above is recorded in our engineering registry with an evidence state and an open engineering state, and that is exactly how they are reproduced here.

If a later edition says any of these is closed, it will say who closed it and with what evidence.

THYLORA / ErsatzReality - C&W Design Notebook No. 001

Source records: er_automotive_gap_registry (ER-AUTO-GAP-001 through 008) and program THY-VEH-SAFE-SERVICEABLE-001.

Design research only. No physical vehicle, no certification, no test data is claimed.